PLUMBING LEGEND

- DOMESTIC COLD WATER PIPING
- DOMESTIC HOT WATER PIPING
- PIPING(UNDER GROUND)
- DOMESTIC HOT WATER RECIRCULATION PIPING
- VENT PIPING
- WASTE (SANITARY SEWER)
- BALANCING VALVE (BV)
- CHECK VALVE
- PRESSURE REDUCING VALVE
- SHUT OFF VALVE
- DIRECTION OF FLOW
- REDUCER OR INCREASER
- TOP CONNECTION, 45 OR 90 DEGREES
- BOTTOM CONNECTION, 45 OR 90 DEGREES
- SIDE CONNECTION
- CAPPED OUTLET
- DROP IN PIPING
- RISE IN PIPING
- UNION
- SOLENOID VALVE
- THERMOSTAT MIXING VALVE
- PRESSURE GAUGE
- VALVE IN RISER
- WATER METER (W)
- STRAINER

PLUMBING SHEET INDEX

SR NO.	SHEET NO.	SHEET NAME	SCALE
1	PG.1	GENERAL NOTES AND SYMBOLS	N.T.S.
2	P1.0	PLUMBING DRAINAGE FLOOR PLAN	1/4" = 1'-0"
3	P2.0	DOMESTIC WATER FLOOR PLAN	1/4" = 1'-0"
4	P3.0	PLUMBING RISER DIAGRAM	N.T.S.
5	P4.0	PLUMBING - DETAILS	N.T.S.
6	P5.0	PLUMBING - SCHEDULES	N.T.S.

PLUMBING ABBREVIATIONS

AAV	AIR ADMITTANCE VALVE
ABV	ABOVE
AFF	ABOVE FINISHED FLOOR
A/E	ARCHITECT/ENGINEER
ACW	AUTOMATIC CLOTHES WASHER
ADA	AMERICANS WITH DISABILITIES ACT
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS
AST	AUTOMATIC SPRAY TAN UNIT
ASTM	AMERICAN SOCIETY FOR TESTING & MATERIALS
BHP	BRAKE HORSEPOWER
BTU	BRITISH THERMAL UNIT
BFP	BACK FLOW PREVENTOR
C/I	CONTRACTOR INSTALLED
CD	CONDENSATE
CONN	CONNECTION
CONT	CONTINUOUS
CV	CHECK VALVE
CW	COLD WATER
DFU	DRAINAGE FIXTURE UNIT
DIA	DIAMETER
DN	DOWN
DWG	DRAWING
ECO	EXTERIOR CLEAN OUT
DW	DISH WASHER
°F	DEGREES FAHRENHEIT
FCO	FLOOR CLEAN OUT
FD	FLOOR DRAIN
GAL	GALLONS
GPM	GALLONS PER MINUTE
HR	HOUR
HW	HOT WATER
HWR	HOT WATER RETURN
HB	HOSE BIB
L	LAVATORY
MX	MIXING VALVE
M.C.	MECHANICAL SUB-CONTRACTOR
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
N.T.S	NOT TO SCALE
PD	PUMP DISCHARGE
PDI	PLUMBING DRAINAGE INSTITUTE
PSIG	POUNDS PER SQUARE INCH
P.C.	PLUMBING SUB-CONTRACTOR
RM	ROOM
SAN	SANITARY
SH	SHOWER
T & P	TEMPERATURE & PRESSURE
TPV	TRAP PRIMER VALVE
TYP	TYPICAL
U/G	UNDERGROUND
V	VENT
W	WASTE
WC	WATER CLOSET
WHA	WATER HAMMER ARRESTOR
WCO	WALL CLEAN OUT
CO	CLEAN OUT
WM	WASHING MACHINE
TW	TEMPERED WATER
WH	WATER HEATER
ET	EXPANSION TANK

REFERENCE CODES

1. 2019 CALIFORNIA BUILDING CODE
2. 2019 CALIFORNIA RESIDENTIAL CODE
3. 2019 CALIFORNIA MECHANICAL CODE
4. 2019 CALIFORNIA ELECTRICAL CODE
5. 2019 CALIFORNIA PLUMBING CODE
6. 2019 CALIFORNIA ENERGY CODE
7. 2019 CALIFORNIA FIRE CODE
8. 2019 CALIFORNIA GREEN BUILDING STANDARDS CODE

EQUIPMENT IDENTIFICATION

XXX — SYMBOL OR TYPE (SEE SCHEDULE OR SPECIFICATIONS)
IDENTIFICATION NUMBER (SEE SCHEDULE)

X / XX — KEYNOTES

PLUMBING SPECIFICATIONS

G E N E R A L

RELATED DOCUMENTS:

1. THE GENERAL REQUIREMENTS OF THE ARCHITECTURAL SPECIFICATIONS ARE PART OF THESE SPECIFICATIONS. WHERE AN INCONSISTENCY EXISTS BETWEEN THE WORDING OR INTENT, THIS SECTION SHALL TAKE PRECEDENCE. THE STANDARD FORM OF "GENERAL CONDITIONS" ISSUED BY THE AMERICAN INSTITUTE OF ARCHITECTS DOCUMENT A201, LATEST EDITION, SHALL FORM PART OF THIS CONTRACT.

2. CONTRACT DOCUMENTS MAY INCLUDE, BUT NOT LIMITED TO THE FOLLOWING: ARCHITECTURAL, STRUCTURAL, ELECTRICAL, MECHANICAL, FIRE PROTECTION, AND CIVIL.

DEFINITIONS:

1. TERMINOLOGY: AS DEFINED IN MSS SP-90, "GUIDELINES ON TERMINOLOGY FOR PIPE HANGERS AND SUPPORTS."

2. WATER DISTRIBUTION PIPING: INTERIOR DOMESTIC WATER PIPING.

3. WATER SERVICE: EXTERIOR DOMESTIC WATER PIPING.

4. ACCESSIBLE FIXTURE: PLUMBING FIXTURE THAT CAN BE APPROACHED, ENTERED, AND USED BY PEOPLE WITH DISABILITIES.

PERFORMANCE REQUIREMENTS:

1. DESIGN SUPPORTS FOR MULTIPLE PIPES CAPABLE OF SUPPORTING COMBINED WEIGHT OF SUPPORTED SYSTEMS, SYSTEM CONTENTS, AND COMPONENTS.

2. DESIGN EQUIPMENT SUPPORTS CAPABLE OF SUPPORTING COMBINED OPERATING WEIGHT OF SUPPORTED EQUIPMENT AND CONNECTED SYSTEMS AND COMPONENTS.

3. DESIGN SEISMIC-RESTRAINT (IF APPLICABLE) HANGERS AND SUPPORTS FOR PIPING AND EQUIPMENT, AND OBTAIN APPROVAL FROM AUTHORITIES HAVING JURISDICTION.

4. COMPONENTS AND INSTALLATION SHALL BE CAPABLE OF WITHSTANDING THE WORKING PRESSURE.

SUBMITTALS:

1. CONTRACTOR SHALL SUBMIT FOR REVIEW TO THE ARCHITECT/ENGINEER A COMPLETE LIST OF ITEMS TO BE FURNISHED AND INSTALLED UNDER THIS CONTRACT. INCLUDING, BUT NOT BE LIMITED TO THE FOLLOWING:

   1.1. DIMENSIONED SHOP DRAWINGS OF MATERIALS, FIXTURES AND EQUIPMENT.
   1.2. DIMENSIONED SHOP DRAWINGS OF EQUIPMENT AND PIPING PLAN LAYOUT(S).
   1.3. PRODUCT DATA.
   1.4. FIELD-QUALITY CONTROL INSPECTION AND TEST REPORTS
   1.5. FIELD TEST CERTIFICATES.
   1.6. OPERATION AND MAINTENANCE DATA.
   1.7. EQUIPMENT MANUALS.
   1.8. VALVE TAGS.
   1.9. WELDING CERTIFICATES.

2. ANY DEFERRED CONTRACTOR SUBMITTALS OR CALCULATIONS REQUIRED BY THE LOCAL AHJ, IN ORDER TO OBTAIN OCCUPANCY SHALL BE PROVIDED THE PLUMBING CONTRACTOR. ALL SUBMITTALS SHALL BE PREPARED BY A LICENSED PROFESSIONAL ENGINEER, LICENSED IN THE PROJECT JURISDICTION, AS REQUIRED BY THE LOCAL AHJ. THE CONTRACTORS ENGINEER SHALL PREPARE PLANS, CUT SHEETS, CALCULATIONS, ETC. AS REQUIRED. ALL DEFERRED SUBMITTALS SHALL BE SIGNED AND STAMPED BY THE PREPARING ENGINEER. THE ELECTRICAL CONTRACTOR SHALL INCLUDE ANY FEES REQUIRED FOR SUCH SUBMISSION IN HIS BID.

CODES AND STANDARDS:

1. APPLICABLE CODES:

   1.1. LOCAL PLUMBING CODE.
   1.2. LOCAL DEPARTMENT OF PUBLIC HEALTH.
   1.3. LOCAL AMENDMENTS TO PLUMBING CODE.
   1.4. LOCAL ORDINANCES.

QUALITY ASSURANCE:

1. WELDING: QUALITY PROCEDURES AND PERSONNEL ACCORDING TO ASME BOILER AND PRESSURE VESSEL CODE: SECTION IX.

2. FIRE TEST RESPONSE CHARACTERISTICS: INSULATION AND RELATED MATERIALS SHALL HAVE FIRE TEST RESPONSE CHARACTERISTICS INDICATED, AS DETERMINED BY TESTING IDENTICAL PRODUCTS PER ASTM E 84, BY A TESTING AND INSPECTING AGENCY ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION. FACTORY LABEL INSULATION AND JACKET MATERIALS AND ADHESIVE, MASTIC, AND CEMENT MATERIAL CONTAINERS, WITH APPROPRIATE MARKINGS OF APPLICABLE TESTING AND INSPECTING AGENCY.

   2.1. INSULATION INSTALLED INDOORS: FLAME SPREAD INDEX OF 25 OR LESS, AND SMOKE DEVELOPED INDEX OF 50 OR LESS.
   2.2. INSULATION INSTALLED OUTDOORS: FLAME SPREAD INDEX OF 75 OR LESS, AND SMOKE DEVELOPED INDEX OF 150 OR LESS.

3. PIPING MATERIALS SHALL BEAR LABEL, STAMP, OR OTHER MARKINGS OF SPECIFIED TESTING AGENCY.

4. ASME COMPLIANCE:

   4.1. ASME B31.9 FOR BUILDING SERVICES PIPING VALVES.
   4.2. ASME COMPLIANCE FOR FERROUS VALVES: ASME B16.10 AND ASME B16.34 FOR DIMENSION AND DESIGN CRITERIA.

5. NSF COMPLIANCE:

   5.1. NSF 14, "PLASTIC PIPING SYSTEM COMPONENTS AND RELATED MATERIALS."
   5.2. NSF 61, "DRINKING WATER SYSTEM COMPONENTS - HEALTH EFFECTS; SECTIONS 1 THROUGH 9."

6. ELECTRICAL COMPONENTS, DEVICES AND ACCESSORIES: LISTED AND LABELED AS DEFINED IN NFPA 70, ARTICLE 100, BY A TESTING AGENCY ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION, AND MARKED FOR INTENDED USE.

7. REGULATORY REQUIREMENTS:

   7.1. COMPLY WITH PUBLIC LAW 101-336, "AMERICAN WITH DISABILITIES ACT," AND "UNIFORM FEDERAL ACCESSIBILITY STANDARDS."
   7.2. COMPLY WITH PUBLIC LAW 102-486, "ENERGY POLICY ACT."

LAWS AND ORDINANCES:

- CONTRACTOR SHALL GIVE ALL NOTICES AND COMPLY WITH ALL LAWS, ORDINANCES, RULES AND REGULATIONS BEARING ON THE CONDUCT OF WORK AS SHOWN ON DRAWINGS AND SPECIFIED HEREIN. IF CONTRACTOR OBSERVES THAT THE DRAWINGS AND SPECIFICATIONS ARE AT VARIANCE THEREWITH, CONTRACTOR SHALL PROMPTLY NOTIFY ARCHITECT/ENGINEER IN WRITING WHEN SUBMITTING BID AND ANY NECESSARY CHANGES SHALL BE ADJUSTED AS PROVIDED IN THE CONTRACT FOR SUCH CHANGES IN WORK. IF CONTRACTOR PERFORMS ANY WORK, CONTRARY TO SUCH LAWS, ORDINANCES, RULES & REGULATIONS, CONTRACTOR SHALL BEAR ALL COSTS FOR CORRECTING THE WORK.

GUARANTEE:

- CONTRACTOR SHALL GUARANTEE WORK TO BE FREE FROM DEFECTIVE WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE (1) YEAR FROM DATE OF FINAL CERTIFICATE. ANY REPAIRS OR REPLACEMENT DURING THIS PERIOD SHALL BE MADE WITHOUT COST TO THE OWNER, UPON OWNER'S REQUEST.

GENERAL NOTES, FIXTURES, & VALUES

- DESIGN AND CONSTRUCT COMPLETE PLUMBING SYSTEM CONSISTING OF DRAIN. VENT WASTE AND WATER LINES IN COMPLIANCE WITH ALL APPLICABLE STATE LOCAL AND FEDERAL LAWS, CODES AND REGULATIONS. COMPLY WITH ARCHITECTURAL WORKING DRAWINGS FOR LOCATIONS OF ALL PLUMBING FIXTURES,TRIM AND RELATED ITEMS.
- PERMITS AND INSPECTIONS,SHALL BE OBTAINED BY THIS CONTRACTOR AS REQUIRED BY LIMITS OF THE WORK DEFINED HEREIN AND BY APPLICABLE CODES AND REGULATIONS.
- DESIGN DRAWINGS SHALL BE PREPARED BY THIS CONTRACTOR IN COMPLIANCE WITH THE PROJECT DESIGN AND IN COMPLIANCE WITH ALL APPLICABLE CODE REQUIREMENTS. SUBMIT PLUMBING DESIGN ISOMETRIC DRAWINGS TO THE BUILDING DEPARTMENT FOR APPROVAL AS PER THEIR REQUIREMENTS.
- PLUMBING FIXTURES AND FITTINGS SHALL BE INSTALLED IN ACCORDANCE WITH THE CALIFORNIA PLUMBING CODES AND SHALL MEET THE APPLICABLE STANDARDS OF THE CALIFORNIA PLUMBING CODE.
- PLUMBING FIXTURES SHALL BE LISTED IN CONFERENCE WITH CPC 2019.
- HOT MOPPED SHOWER PAN SHALL BE INSPECTED UPON COMPLETION OF HOT-MOPPING AND SHALL BE FILLED WITH WATER FOR INSPECTION. CPC 2019.CIRCULATING HOT WATER SYSTEMS SHALL BE FULLY INSULATED AND SHALL HAVE CONTROL (TIME CLOCK) CAPABLE OF AUTOMATICALLY TURNING OFF THE CIRCULATING PUMP(S) WHEN HOT WATER IS NOT REQUIRED. TITLE 24.
- WATER CLOSETS, URINALS,SHOWER HEADS, LAVATORY FAUCETS AND SINK FAUCETS SHALL MEET THE FLOWING REQUIREMENTS OUTLINED IN THE TITLE 24 "APPLIANCE EFFICIENCY STANDARDS".

WATER CLOSETS	1.28 GPF
URINALS	0.5 GPF
SHOWER HEADS	1.8 GPM
LAVATORIES	1.2 GPM
SINKS	1.8 GPM

- FOR ADDITIONS OR IMPROVEMENTS TO A RESIDENCE BUILT BEFORE 1994,EXISTING "NONCOMPLIANT" FIXTURES (TOILETS THAT USE MORE THAN 1.6 GALLONS OF WATER PER FLUSH, URINALS THAT USE MORE THAN ONE GALLON OF WATER PER FLUSH, SHOWERHEADS THAT HAVE A FLOW CAPACITY OF MORE THAN 2.5 GALLONS OF WATER PER MINUTE,AND INTERIOR FAUCETS THAT EMIT MORE THAN 2.2 GALLONS OF WATER PER MINUTE) SHALL BE REPLACED. CERTIFICATION OF COMPLIANCE SHALL BE GIVEN TO THE BUILDING INSPECTOR PRIOR TO FINAL PERMIT APPROVAL. CALIFORNIA SB407.
- SHOWER AND TUB/SHOWER CONTROL VALVES SHALL BE OF THE PRESSURE BALANCE OR THERMOSTATIC MIXING TYPE. HANDLE POSITION SPOTS SHALL BE PROVIDED ON THESE VALVES AND SHALL BE ADJUSTED TO DELIVER A MAXIMUM WATER SETTING OF 120 AS PER CPC 2019.
- PROVIDE ALL VALVES AS REQUIRED. SWEAT COPPER ENDS ON COPPER PIPING SYSTEMS,SCREWED ENDS ON STEEL SYSTEMS. USE SQUARE HEAD COCKS ON FUEL. GAS SYSTEMS.PROVIDE SPEEDWAY STOPS ON ALL FIXTURES.
- WATER PRESSURE REGULATORS SHALL BE INSTALLED ON THE WATER SERVICE PRIOR TO CONNECTION TO BUILDING IF PRESSURE EXCEEDS 80 PSI AS PER CPC 2019.
- HANGERS AND SUPPORTS SHALL BE BY A RECOGNIZED MANUFACTURER SPECIALIZING IN THESE PRODUCTS, SUCH AS MCO, BLINE, ETC
- IN SHOWERS & TUB-SHOWER COMBINATIONS , CONTROL VALVES MUST BE PRESSURE BALANCED OR THERMOSTATIC MIXING VALVES AS PER CPC 2019.
- WATER SYSTEM SHALL BE STERILIZED USING CHLORINE SOLUTION, TESTED AND CERTIFICATION PROVIDED FROM APPROVED TESTING FACILITY.
- ALL SYSTEMS SHALL BE TESTED IN ACCORDANCE WITH THE APPLICABLE SECTIONS OF CALIFORNIA BUILDING CODE AND WITNESSED BY THE LOCAL BUILDING AUTHORITY AND OWNERS REPRESENTATIVE.
- ALL HOSE BIBBS AND FAUCETS FITTED WITH THREADED CONNECTIONS SHALL BE FITTED WITH AN APPROVED NON-REMOVABLE BACK FLOW DEVICE PER CPC 2019
- BUILT-UP SHOWER RECEPTORS SHALL BE INSTALLED AND INSPECTED UPON COMPLETION OF WATER PROOFING AS PER CPC 2019.
- COUNTERSINK PLUG TO BE INSTALLED WHERE RAISED HEADS CAUSE A HAZARD.
- CONDENSATE SHALL BE COLLECTED AND DISCHARGED TO AN APPROVE PLUMBING FIXTURES OR APPROVED DISPOSAL AREA. REFER TO UMC SECTION 310 FOR ADDITIONAL REQUIREMENTS FOR CONDENSATE DRAINAGE PIPING.
- APPROVED SPARK ARRESTORS SHALL BE INSTALLED ON ALL CHIMNEYS AS PER CBC 2019.

WATER HEATERS:

- THE WATER HEATERS TO BE INSTANTANEOUS TYPE – SEE TITLE 24 CALCS.
- WATER HEATERS SHALL BE EQUIPPED WITH A COMBINATION PRESSURE AND TEMPERATURE RELIEF VALVE AT ALL WATER HEATERS, SET TO OPEN AT NOT MORE THAN 150 PSI. DRAIN PIPE SHALL EXTEND TO OUTSIDE OF BUILDING AND TERMINATE NOT MORE THAN 2 FEET NOT LESS THAN 6 INCHES ABOVE THE GROUND AND POINT DOWNWARD WITH THE END UNOBSTRUCTED. ANY OTHER TERMINATION LOCATION SHALL REQUIRE APPROVAL BY THE BUILDING OFFICIAL AS PER CPC 2019.
- DOMESTIC WATER HEATERS SHALL BE CERTIFIED AND LISTED BY THE CALIFORNIA ENERGY COMMISSION. EXTERIOR INSULATION SHALL BE R-16, OR GREATER. THE FIRST 5 FEET OF EXPOSED HOT AND COLD WATER PIPING AND THE WATER HEATER SHALL BE INSULATED. TITLE 24.
- PROVIDE A R-12 WATER HEATER BLANKET FOR ALL WATER HEATERS.
- WATER HEATERS LOCATED IN ATTIC OR FURRED SPACE WHERE DAMAGE MAY RESULT FROM LEAKING WATER HEATERS SHALL BE INSTALLED IN OR OVER A WATER TIGHT PAN WITH A 3/4 INCH DRAIN TO AN APPROVED LOCATION AS PER CPC 2019.
- IN SEISMIC DESIGN CATEGORIES D & E WATER HEATER SHALL BE ANCHORED OR STRAPPING SHALL BE AT POINT WITHIN THE UPPER ONE-THIRD (1/3) AND LOWE ONE-THIRD (1/3) OR ITS VERTICAL DIMENSIONS AT THE LOWER POINT, A MINIMUM DISTANCE OF FOUR (4) INCHES SHALL BE MAINTAINED ABOVE THE CONTROLS WITH THE STRIPPING. (CPC 2019).
- ALL HOT WATER PIPES RUN FROM THE HEATING SOURCE TO THE KITCHEN FIXTURES SHALL INCLUDE A MINIMUM OF 1 INCH INSULATION." (CALIFORNIA ENERGY CODE)
- PROVIDE MINIMUM R-12 INSULATION FOR SMALL STORAGE WATER HEATERS WITH AN INPUT RATING LESS THAN OR EQUAL TO 75,000 BTU/HR. (CALIFORNIA ENERGY CODE)
- THE ENTIRE LENGTH OF RE-CIRCULATING DISTRIBUTION SECTIONS OF DOMESTIC WATER SYSTEMS SHALL BE INSULATED, REGARDLESS OF WHETHER THE PIPING IS BURIED OR EXPOSED." (CALIFORNIA ENERGY CODE)
- REQUIREMENT FOR GAS WATER HEATER INSTALLED TO SERVE INDIVIDUAL DWELLING UNITS ES150.0(N)

a. A CONDENSATE DRAIN INSTALLED NO HIGHER THAN 2" ABOVE THE BASE OF THE HEATER THAT ALSO ALLOWS FOR GRAVITY DRAINAGE
b. THE "B" VENT INSTALLED IN A STRAIGHT POSITION (NO BENDS) FROM THE ROOM CONTAINING THE WATER HEATS TO THE ROOF TERMINATION (FOR FUTURE POSSIBLE SLEEVING FOR HIGH EFFICIENCY HEATER VENTING).
c. A 120 VOLT RECEPTACLE ACCESSIBLE TO THE HEATER INSTALLED WITH TO 3'-0".

PIPING:

- INSTALLATION OF ALL PIPING SYSTEMS SHALL BE IN ACCORDANCE WITH APPLICABLE LAPMO INSTALLATION STANDARDS AND LISTED IN CPC APPENDIX 1.
- SOLDER SHALL BE LEAD FREE COMPLYING WITH THE PROVISIONS OF FEDERAL CLEAN DRINKING WATER ACT AND THE CPC 2019.
- 50/50 SOLDER JOINTS AREA NO LONGER PERMITTED IN PORTABLE WATER DISTRIBUTION SYSTEMS. ONLY 95/5 SOLDER JOINTS ARE PERMISSIBLE.
- INFORMATIONAL,FOR THE PURPOSES OF THIS SECTION ,"LEAD FREE" MEANS NOT MORE THAN 0.2 PERCENT LEAD WHEN USED WITH RESPECT TO SOLDER AND FLUX AND NOT MORE THAN 8PERCENT WHEN USED WITH RESPECT TO PIPES AND PIPE FITTINGS WITH RESPECT TO PLUMBING FITTING AND FIXTURES, "LEAD FREE" MEANS NOT MORE THAN 4 PERCENT BY DRY WEIGHT.
- SUPPORT ALL LINES BY MEANS OF RESILENT FITTING OR ACCESSORIES, DO NOT ALLOW PIPING TO BE IN DIRECT CONTACT WITH FRAMING.
- NO COPPER PIPING SHALL BE IN CONTACT WITH ANY METAL OTHER THAN COPPER.
- PIPING SYSTEMS SHALL BE SECURELY SUPPORTED TO PREVENT UNDUE MOVEMENT AND SWAYING BUT FLEXIBLE ENOUGH TO PERMIT EXPANSION.
- ALL DOMESTIC HOT WATER PIPING TO HAVE THE FOLLOWING MINIMUM INSULATION INSTALLED: PIPE(1 INSULATION), 1"PIPE (1" INSULATION); 1/2" PIPE (1/2" INSULATION) 3/4"PIPE (1" INSULATION)1" TO 1-1/2" PIPE (1-1/2" INSULATION).(CPC 2019)
- ADDITIONALLY,THE 1/2" HOT WATER PIPE TO THE KITCHEN SINKED THE COLD WATER PIPE WITHIN 5' OF THE WATER BOTH REQUIRE 1" MINIMUM ISULATION ES 150.0(J).
- BELOW GRADE HOT WATER PIPING IS REQUIRED TO BE INSTALLED IN A WATERPROOF AND NON -CRUSHABLE SLEEVE OR CASING THAT ALLOWS FOR REPLACEMENT OF BOTH THE PIPING AND INSULATION PLEASE NOTE OR SHOW THIS ON THE PLANS.

- DRAIN PIPING SHALL SLOPE AT 2% OR 1/4 INCHES PER FOOT FOR BELOW 4" PIPING UNLESS OTHERWISE NOTED. 4' AND LARGER PIPING REQUIRING SLOPE 1% OR 1/8 INCHES PER FOOT. THE INSTALLATION CONDITIONS SHALL BE NOTED ON THE PLANS AND APPROVAL RECEIVED FROM THE ADMINISTRATIVE AUTHORITY PER CPC 2019.
- COMPENSATE SHALL BE COLLECTED AMP DISCHARGED TO AN APPROVED DISPOSAL AREA. REFER TO CPC 2019 FOR ADDITIONAL REQUIREMENTS FOR COMPENSATE DRAINAGE PIPING.
- F.M.C.O. SHALL OPEN IN A DIRECTION OPPOSITE TO THE FLOW OF THE WASTE ABOVE PIPE F.L
- PROVIDE MINIMUM CLEARANCE OF NOT LESS THAN 18" IN FRONT OF CLEAN OUT.
- ALL SOIL WASTE AND VENT LINES SHALL COMPLY WITH UNIFORM PLUMBING CODE. SOIL AND WASTE BEYOND THE FIVE FOOT BUILDING LINE SHALL BE VITRIFIED CLAY PIPE WITH SPEED SEAL JOINTS OR TRANSIT SEWER PIPE WITH RING GROOVE COUPLINGS.
- COLD AND HOT WATER LINES ABOVE GRADE SHALL BE TYPE M HARD DRAWN COPPER WITH 50-50 SOLDER JOINTS. COLD WATER LINES BELOW GRADE SHALL BE TYPE L COPPER WITH 95-5 SOLDER JOINTS FROM WATER METER.
- FLOOR DRAINS AND FLOOR SINKS WHICH ARE SUBJECT TO INFREQUENT USE SHALL BE PROVIDED WITH AN APPROVED AUTOMATIC METHOD OF MAINTAINING TRAP SEALS AS PER CPC 2019.
- A MINIMUM OF 12" ACCESS PANEL TO BATHTUB TRAP CONNECTIONS SHALL BE INSTALLED WHEN SLIP JOINT CONNECTIONS ARE USED AS PER CPC 2019.
- NO GAS PIPING SHALL BE INSTALLED UNDERGROUND BENEATH ANY BUILDING SLAB ON GRADE OR STRUCTURE UNLESS INSTALLED IN A GAS TIGHT CONDUIT. INCLUDING PIPING UNDER COVERED PATIOS [CPC 2019]
- NO GAS PIPE SHALL BE INSTALLED UNDER THE BUILDING O PATIO COVER
- UNDERGROUND PLASTIC GAS PIPING SHALL BE 18" DEEP AND THE TRANSITION RISER MUST COMPLY WITH CPC 2019.
- AN ELECTRICALLY CONTINUOUS CORROSION-RESISTANT TRACER WIRE (MINIMUM AWG 14) OR TAE SHALL BE BURIED WITH THE PLASTIC PIPE TO FACILITATE LOCATING. ONE (1) END SHALL BE BROUGHT ABOVE GROUND AT A BUILDING WALL OR RISER. [NFPA 54:1.133] GPG 1211.13

VENTS:

- VENT TERMINATION SHALL BE NOT LESS THAN 10 ' FEET FROM OR AT LEAST 3 ' FEET ABOVE AIR INLET PER CPC 2019.
- PROVIDE DRYER VENT TO OUTSIDE PER UMC.
- GAS VENTS AND NONCOMBUSTIBLE PIPING IN WALLS,PASSING THROUGH THREE FLOORS OR LESS, SHALL BE EFFECTIVELY DRAFT STOPPED AT EACH FLOOR OR CEILING.(CBC 2019).
- PROVIDE A SEPARATE VENT FOR EACH WASTE BRANCH LINE EXCEEDING 15 FEET IN LENGTH.
- THE MINIMUM AREA OF ANY VENT INSTALLED IN A COMBINATION WASTE AND VENT SYSTEM SHALL BE AT LEAST 2' GROSS SECTIONAL AREA OF THE DRAINPIPE SERVED.

E X E C U T I O N

SCOPE OF WORK:

1. PROVIDE ALL LABOR, MATERIAL, EQUIPMENT, FACILITIES, TRANSPORTATION, FEES AND SERVICES NECESSARY FOR A COMPLETE PLUMBING SYSTEM(S) AS INDICATED ON THE DRAWINGS AND SPECIFIED HEREIN. WORKMANSHIP SHALL BE COMPLETE IN EVERY ASPECT, TESTED, APPROVED AND SATISFACTORY TO THE ARCHITECT/ENGINEER AND IN ACCORDANCE WITH LOCAL, STATE AND FEDERAL LAWS HAVING JURISDICTION.

2. IT IS THE DECLARED AND ACKNOWLEDGED INTENT OF THESE SPECIFICATIONS TO PROVIDE A COMPLETE PLUMBING SYSTEM(S), INCLUSIVE OF ALL REQUIRED PARTS AND ACCESSORIES COMPLETE AND READY FOR USE AS DESCRIBED, BUT NOT LIMITED TO THE FOLLOWING:

   2.1. DOMESTIC WATER SERVICE AND DISTRIBUTION.
   2.2. DOMESTIC HOT WATER.
   2.3. SANITARY DRAINAGE SYSTEM.
   2.4. PLUMBING FIXTURES.
   2.5. PLUMBING EQUIPMENT.
   2.6. PLUMBING SPECIALTIES.

3. THE DRAWINGS INDICATE DIAGRAMMATICALLY THE EXTENT AND LOCATION OF THE WORK INCLUDED. WORK INDICATED, BUT HAVING MINOR DETAILS OBVIOUSLY OMITTED, SHALL BE PROVIDED, INCLUDING THESE DETAILS, WITHOUT EXTRA COST TO THE OWNER.

DISINFECTION OF POTABLE WATER SYSTEM:

1. DISINFECT AND FLUSH POTABLE WATER SYSTEM PER LOCAL PLUMBING, BUILDING, AND HEALTH DEPARTMENT REQUIREMENTS.

FLUSHING:

1. CHLORINATED WATER SUPPLY: IF WATER SUPPLY SERVING SYSTEM IS CHLORINATED FROM COMMUNITY WATER SYSTEM. SYSTEM SHALL BE FLUSHED WITH CLEAN, POTABLE WATER UNTIL NO DIRTY WATER APPEARS AT THE POINT OF OUTLET.

2. NON-CHLORINATED WATER SUPPLY: WATER SUPPLY SYSTEM SHALL BE FLUSHED WITH CLEAN, POTABLE WATER UNTIL NO DIRTY WATER APPEARS AT THE POINT OF OUTLET.

LABEL AND IDENTIFICATION:

1. IDENTIFICATION FOR ALL PIPING SYSTEM(S) SHALL COMPLY WITH ANSI A13.1 FOR SIZE OF LETTERING AND BACKGROUND COLOR FIELD.

2. PIPING SYSTEM(S): IDENTIFICATION SHALL INCLUDE THE CONTENTS OF THE PIPING SYSTEM(S) AND AN ARROW INDICATING THE DIRECTION OF FLOW. HAZARDOUS PIPING SYSTEM(S) SHALL ALSO CONTAIN INFORMATION ADDRESSING THE NATURE OF THE HAZARD. IDENTIFICATION SHALL BE REPEATED AT MAXIMUM INTERVALS OF TWENTY-FIVE (25) FEET AND AT EACH POINT WHERE PIPING PASSES THROUGH A WALL, FLOOR OR ROOF. COLOR OF THE PIPE IDENTIFICATION SHALL BE DISCERNIBLE AND CONSISTENT THROUGHOUT THE BUILDING.

3. EQUIPMENT: IDENTIFICATION SHALL INCLUDE SYSTEM NUMBER, CAPACITY, FLOW, RATE, STATIC PRESSURE, PUMP HEAD, HORSEPOWER, VOLTAGE, ETC.

4. VALVE TAGS: PROVIDE BRASS VALVE TAGS AND BRASS "S" HOOK FASTENERS WITH VALVE NUMBER AND TYPE OF SERVICE NOTED ON TAG. PROVIDE DUPLICATE CHARTS, THE CHART SHALL BE FOR ALL VALVES AND SHALL INDICATE VALVE IDENTIFICATION NUMBER, LOCATION AND PURPOSE.

RECORD DRAWING SUBMITTALS:

AT PROJECT CLOSE-OUT, CONTRACTOR SHALL SUBMIT RECORD DRAWINGS (CERTIFIED OR APPROVED) ALSO KNOWN AS "AS-BUILT" DRAWINGS. LAYOUT SUBMITTALS SHALL BE SUBMITTED FOR RECORDS PRIOR TO FINAL ACCEPTANCE IN ELECTRONIC FORM ON COMPACT DISC(S), USING AUTOCAD VERSION 2010 "DWG" FORMAT OR ADOBE ACROBAT "PDF" FORMAT VIEWABLE FROM ADOBE. RECORD DATA EQUIPMENT, FIXTURE AND RELATED APPURTENANCES MAY BE SUBMITTED AS BOUND HARDCOPY OR ELECTRONIC ADOBE ACROBAT "PDF" FORMAT.

2. PROVIDE RECORD SUBMITTALS AS FOLLOWS:
   2.1. OWNER: 1 COPY.
   2.2. ARCHITECT OF RECORD: 1 COPY.
   2.3. ENGINEER OF RECORD: 1 COPY.

3. CONTRACTOR IS NOT ALLOWED TO USE THE CONTRACT DOCUMENTS FOR "AS-BUILT" DRAWINGS. BACKGROUNDS SHALL BE IN THE LATEST RELEASE OF AUTOCAD.

RESIDENCE OF
JOE HUBBARD
Project

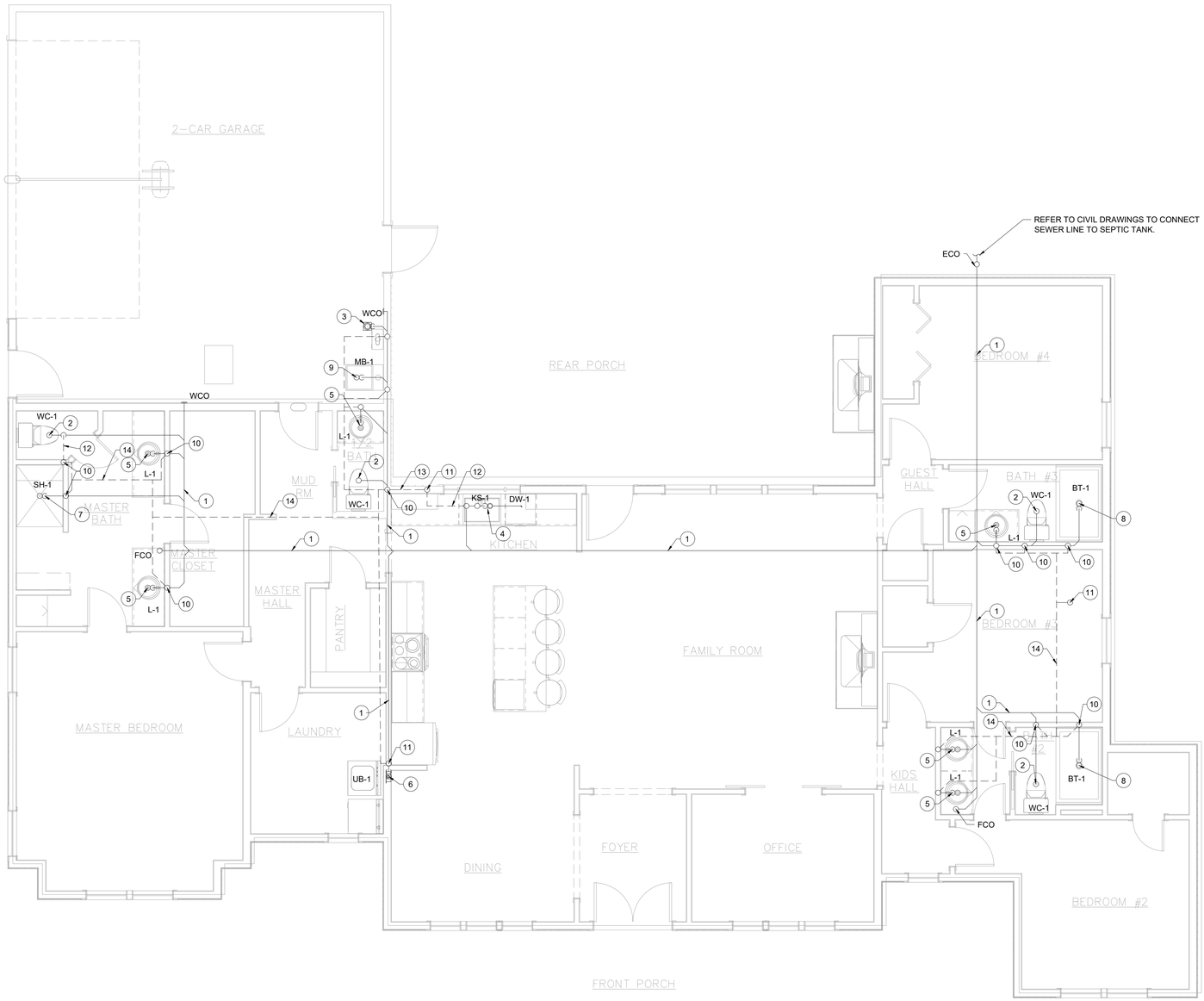
3495
COUNTY ROAD P
GLENN COUNTY,CA

Project No.:
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Sheet Title
GENERAL NOTES AND SYMBOLS

☐ Preliminary Dwg.
☐ Bidding Doc.
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Sheet:
PG.1



1 P1.0 PLUMBING DRAINAGE FLOOR PLAN

SCALE: 1/4" = 1'-0"

NORTH

PLUMBING GENERAL NOTES :

1. PROVIDE FIRE BLOCKING AT OPENING AROUND VENTS, AND PIPES AT FIRE WALL AND FLOOR LEVEL WITH AN APPROVED MATERIAL TO RESIST THE FREE PASSAGE OF FLAME AND PRODUCTS OF COMBUSTION.
2. ALL SANITARY WASTE PIPE RUN BELOW THE FLOOR.

KEY NOTES

1. 4"Ø SAN PIPE.
2. 3"Ø SAN PIPE UP TO WC-1.
3. 2"Ø SAN PIPE UP TO FD-1.
4. 2"Ø SAN UP TO KS-1.
5. 2"Ø SAN UP TO L-1.
6. 2"Ø SAN UP TO UB-1
7. 2"Ø SAN UP TO SH-1
8. 3"Ø SAN PIPE UP TO BT-1.
9. 3"Ø SAN PIPE UP TO MB-1.
10. 2"Ø VENT PIPE UP.
11. 2 1/2"Ø VENT PIPE UP & 3"Ø VTR.
12. 2"Ø VENT PIPE RUN UNDERGROUND.
13. 2 1/2"Ø VENT PIPE.
14. 2"Ø VENT PIPE.

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GLENN COUNTY,CA

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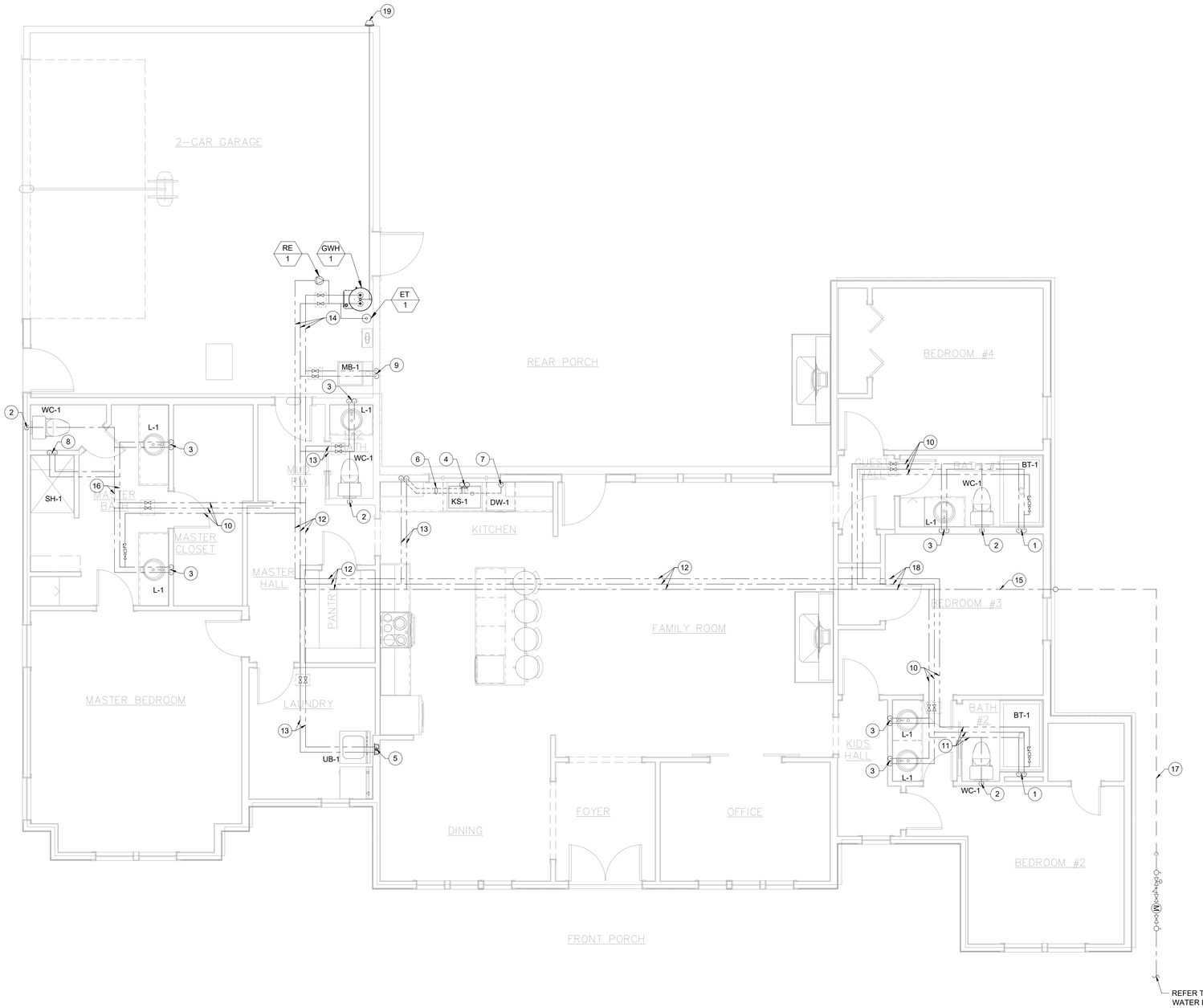
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PLUMBING DRAINAGE FLOOR PLAN

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Sheet:

P1.0



1 P2.0 DOMESTIC WATER FLOOR PLAN

SCALE: 1/4" = 1'-0"

NORTH

PLUMBING GENERAL NOTES :

1. FOR CLARITY, NOT ALL VALVES HAVE BEEN SHOWN. PROVIDE SHUTOFF VALVES IN DOMESTIC WATER PIPING SERVING EACH ROOM WITH FIXTURES. ANGLE STOPS SHALL NOT BE CONSIDERED SHUTOFF VALVES.
2. REFER TO THE PLUMBING ROUGH-IN SCHEDULES FOR THE SIZES OF BRANCH PIPES OF FIXTURES NOT SHOWN ON PLANS.
3. REFERS TO FIXTURE & EQUIPMENT DESIGNATION. SEE CORRESPONDING SPECIFICATION AND EQUIPMENT SCHEDULE FOR FURTHER INFORMATION.
4. PROVIDE ACCESS DOORS FOR SHOCK ABSORBERS, TRAP PRIMERS AND VALVES. COORDINATE FINAL LOCATIONS WITH PARTITIONS AND ARCHITECT.

KEY NOTES:

1. 3/4"Ø CW, 3/4"Ø HW PIPE UP TO BT-1.
2. 1/2"Ø CW PIPE UP TO WC-1.
3. 1/2"Ø CW PIPE, 1/2"Ø HW UP TO L-1
4. 1/2"Ø CW PIPE, 1/2"Ø HW PIPE PIPE UP TO KS-1.
5. 1/2"Ø CW PIPE, & 1/2"Ø HW PIPE UP TO UB-1.
6. 1/2"Ø CW PIPE, & 1/2"Ø HW PIPE RUN UNDER FLOOR.
7. 1/2"Ø HW PIPE PIPE UP TO DW-1.
8. 1/2"Ø CW PIPE, 1/2"Ø HW UP TO SH-1.
9. 1/2"Ø CW, & 1/2"Ø HW PIPE UP TO MB-1.
10. 1"Ø CW, 3/4"Ø HW PIPE, 1/2"Ø HWR PIPE.
11. 3/4"Ø CW, 1/2"Ø HW PIPE & 1/2"Ø HWR PIPE.
12. 1 1/4"Ø CW, 1"Ø HW PIPE, 3/4"Ø HWR PIPE.
13. 1/2"Ø CW, 1/2"Ø HW PIPE.
14. 1 1/4"Ø CW, 1 1/4"Ø HW PIPE, 3/4"Ø HWR PIPE.
15. 1 1/2"Ø CW PIPE.
16. 3/4"Ø CW PIPE, 1/2"Ø HW PIPE.
17. 1 1/2"Ø CW PIPE UNDER GROUND.
18. 1 1/4"Ø CW, 3/4"Ø HW PIPE & 1/2"Ø HWR PIPE.
19. COMBINATION FLUE VENT THROUGH WALL. INSTALLATION AS PER MANUFACTURER'S INSTRUCTION.

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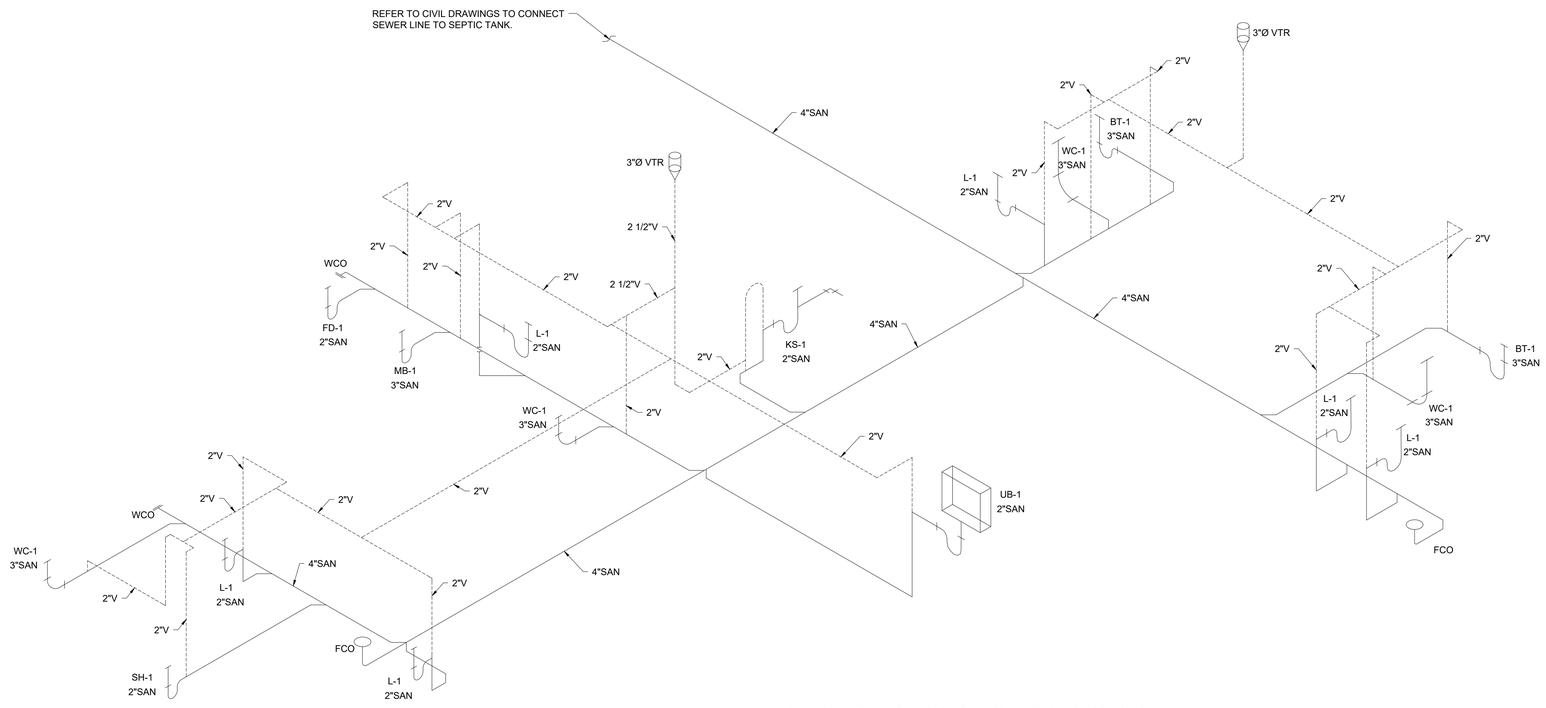
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DOMESTIC WATER FLOOR PLAN

☐ Preliminary Dwg.
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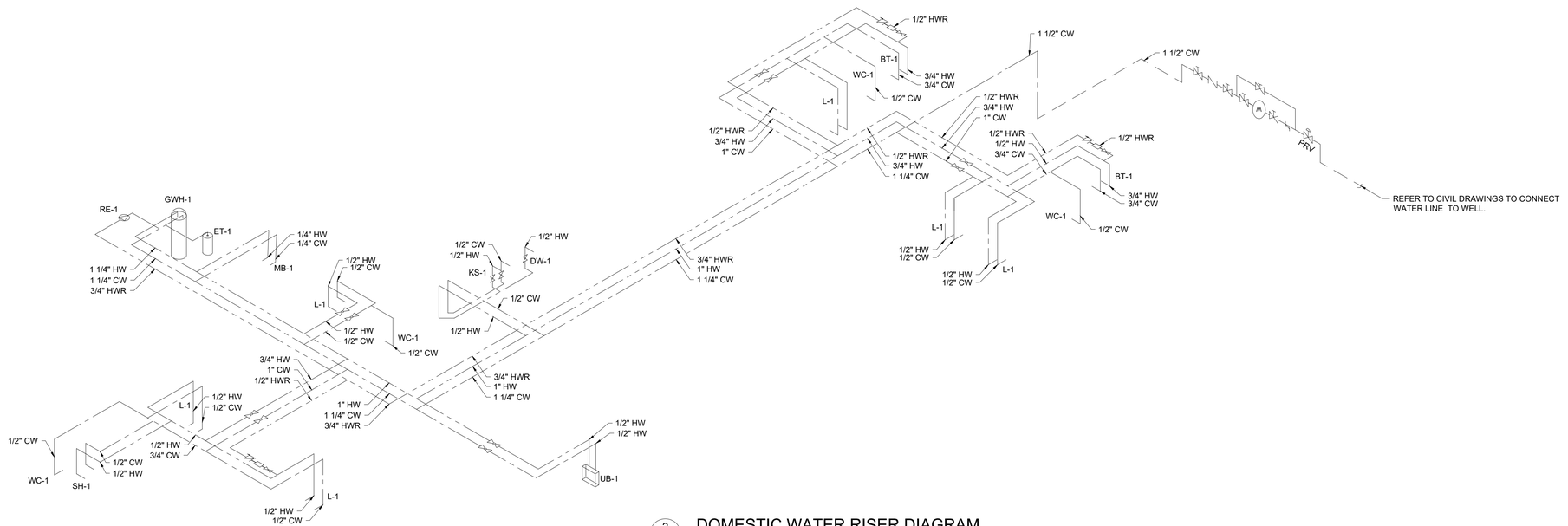
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1 P3.0 SANITARY DRAINAGE RISER DIAGRAM

SCALE: N.T.S.



2 P3.0 DOMESTIC WATER RISER DIAGRAM

SCALE: N.T.S.

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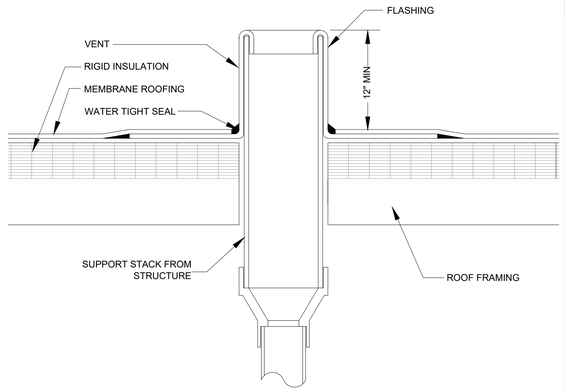
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PLUMBING RISER DIAGRAM

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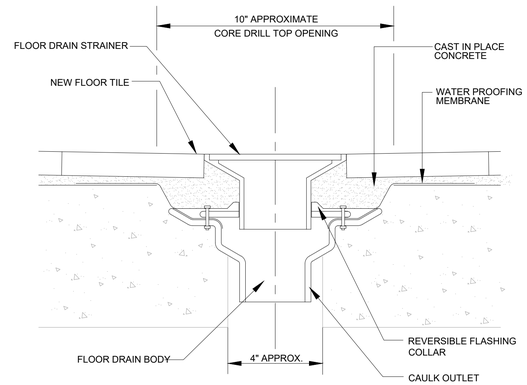
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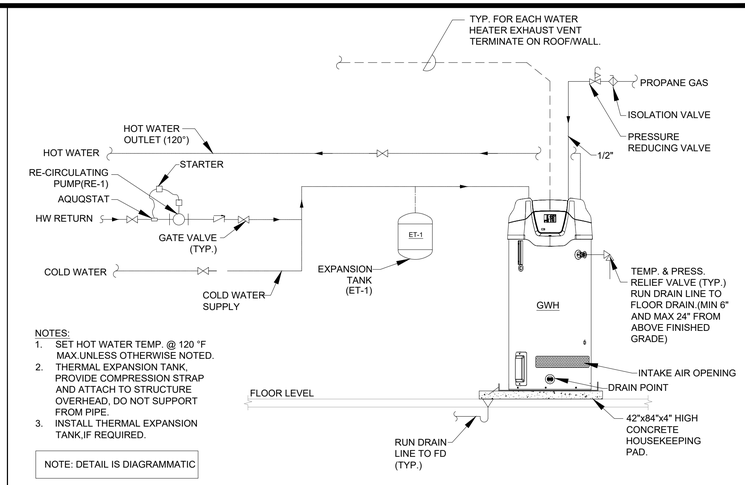
VENT THROUGH ROOF DETAIL 3

NOT TO SCALE



FLOOR DRAIN DETAIL 2

NOT TO SCALE

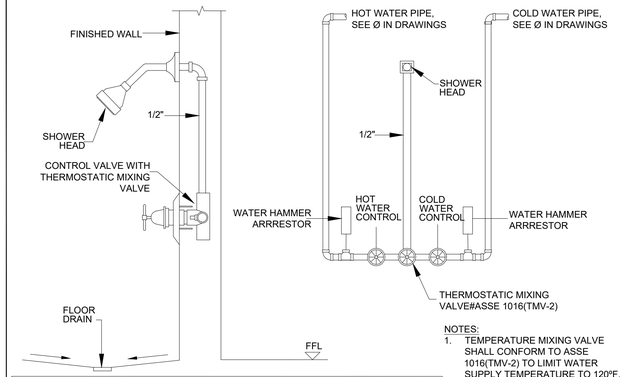


NOTES:
1. SET HOT WATER TEMP. @ 120 °F MAX.UNLESS OTHERWISE NOTED.
2. THERMAL EXPANSION TANK, PROVIDE COMPRESSION STRAP AND ATTACH TO STRUCTURE OVERHEAD, DO NOT SUPPORT FROM PIPE.
3. INSTALL THERMAL EXPANSION TANK,IF REQUIRED.

NOTE: DETAIL IS DIAGRAMMATIC

GAS WATER HEATER CONNECTION DETAIL 1

NOT TO SCALE

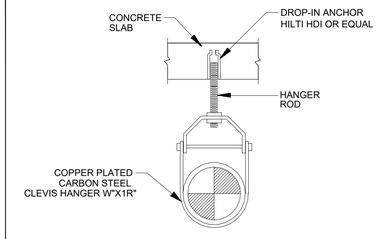


NOTES:
1. TEMPERATURE MIXING VALVE SHALL CONFORM TO ASSE 1016(TMV-2) TO LIMIT WATER SUPPLY TEMPERATURE TO 120ºF.
2. TEMPERATURE MIXING VALVE SHALL BE PRESSURE INDEPENDENT TYPE.

NOTES:
1 VERIFY LOCATION OF DRAIN AND PIPES WITH ARCHITECTURAL DRAWINGS.
2 PIPES SHALL NOT BE INSIDE A STRUCTURAL WALL.
3 ALL EXPOSED PIPES AND FITTINGS SHALL BE CHROME-PLATED.
4 CHROME-PLATED ESCUTCHEON SHALL BE INSTALLED AT EVERY PIPE THAT PASSED THRU A WALL.

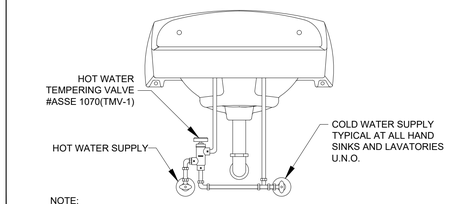
TYPICAL WATER SHOWER DETAILS 10

NOT TO SCALE



PIPE HANGER SUPPORT DETAIL 6

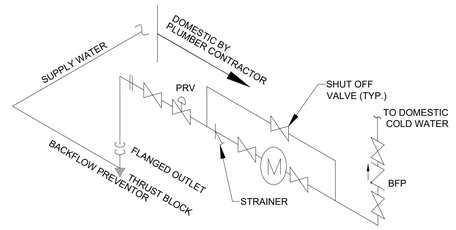
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NOTE:
1. PROVIDE TRAP AND SUPPLY INSULATION AS REQUIRED. SEE SPECIFICATIONS.
2. IF HOT WATER IS SUPPLIED AT 110°F,WE DO NOT NEED TEMP VALVE.

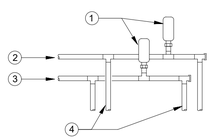
LAVATORY AND HAND SINK TEMPERING VALVE DETAIL 5

NOT TO SCALE



DOMESTIC WATER METERING STATION DETAIL 4

NOT TO SCALE

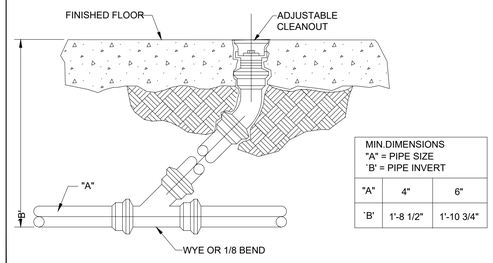


(#) KEY NOTES:
1. WATER HAMMER ARRESTOR (WHA) INSTALL AHEAD OF LAST FIXTURE SERVED BY BRANCH. (TYP)
2. CW PIPE
3. HW PIPE
4. CONNECT TO FIXTURE

WATER ARRESTOR SCHEDULE		
P.I.D. SYMBOL	FIXTURE UNIT RATING	SIZE (MM)
A	1-11	25
B	12-32	32
C	33-60	40
D	61-113	50

WATER HAMMER ARRESTER DETAIL 9

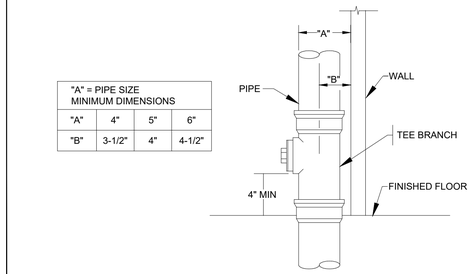
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MIN.DIMENSIONS "A" = PIPE SIZE `B' = PIPE INVERT		
"A"	4"	6"
`B'	1'-8 1/2"	1'-10 3/4"

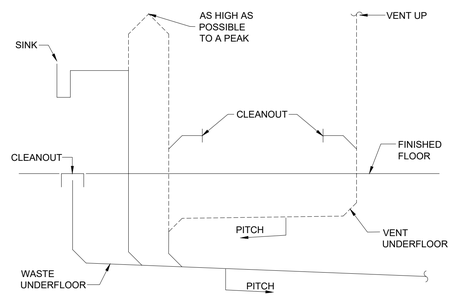
TYPICAL CLEANOUT TO FINISHED FLOOR 8

NOT TO SCALE



CLEANOUT AT BASE OF RISER DETAIL 7

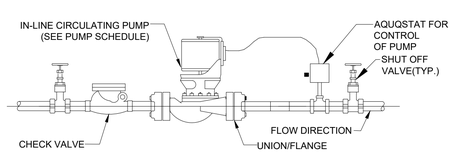
NOT TO SCALE



NOTES:
1. PROVIDE DRAINAGE FITTINGS ON ALL SECTIONS OF THE VENT UNDER FLOOR AND PROVIDE MINIMUM SLOPE OF 1/4" PER FOOT TOWARD DRAINAGE POINT.

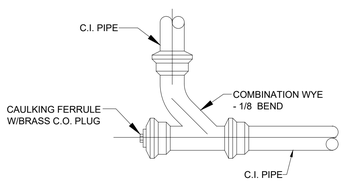
ISLAND VENTING DETAILS 14

NOT TO SCALE



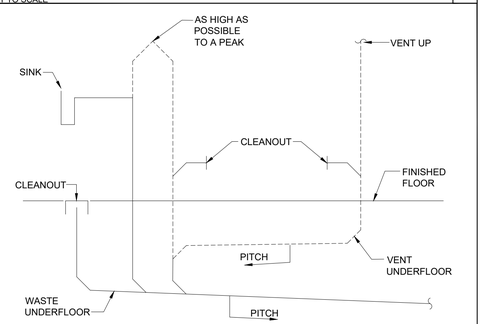
HOT WATER RECIRCULATION PUMP DETAILS 13

NOT TO SCALE



EXPOSED CLEANOUT DETAILS 12

NOT TO SCALE



NOTES:
1. PROVIDE DRAINAGE FITTINGS ON ALL SECTIONS OF THE VENT UNDER FLOOR AND PROVIDE MINIMUM SLOPE OF 1/4" PER FOOT TOWARD DRAINAGE POINT.

ISLAND VENTING DETAILS 11

NOT TO SCALE

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Project

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PLUMBING DETAILS

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PLUMBING FIXTURE SCHEDULE

SYMBOL	MANUFACTURER	MODEL	FIXTURE	MOUNTING	HW	CW	WASTE	VENT	ACCESSORIES / REMARKS	
BFP-1	WATTS	LF007	-	-	-	-	-	-	1 1/2" RPZ, TEMPERATURE RANGE:33°F-180°F, MAXIMUM WORKING PRESSURE:175 PSI, LEAD FREE.	
ECO	ZURN	Z-1400-S	EXTERNAL CLEAN OUT	FLOOR	-	-	SEE REMARKS	-	MATCH CONNECTED PIPE SIZE UP TO 4". CAST IRON BODY WITH BOTTOM OUTLET.	
L-1	KOHLER	K-2882	LAVATORY	WALL	1/2"	1/2"	2"	2"	VITREOUS CHINA, UNDER MOUNT, ASME A112.19.2/CSA B45.1, ICC/ANSI A117.1, 17-1/4"(LENGTH)X 13"(WIDTH) X 3-1/8"(WATER DEPTH).	
BT-1	KOHLER	K-710-W	BATH TAB	FLOOR	3/4"	3/4"	3"	2"	66"X36" FREESTANDING OVAL BATH, OVAL BASIN CREATES A SPACIOUS BATHING EXPERIENCE.	
WC-1	KOHLER	K3999	WATER CLOSET	FLOOR	-	3/4"	3"	2"	WATER CLOSET: ELONGATED BOWL, TWO-PIECE DESIGN, SINGLE-FLUSH GRAVITY FORCE. 1.28 GALLON FLUSH.	
WHA	PRECISION PLUMBING PRODUCTS	SC SERIES	WATER HAMMER ARRESTOR	-	-	-	-	-	INSTALL AND SIZE PER MANUFACTURER'S INSTRUCTIONS. INSTALL AT ENDS OF COLD AND HOT WATER MAIN BRANCHES AND AT FAST-CLOSING VALVES OR ONE AT EACH FIXTURE. SHOW LOCATIONS ON RECORD DRAWINGS.	
BALL VALVES	APOLLO / COBRANCO OR EQUAL	-	BALL VALVES	-	-	-	-	-	FULL PORT, BRASS BODY, 600 WOG	
PIPE HANGERS	B-LINE OR EQUAL	200F	PIPE HANGERS	-	-	-	-	-	-	
KS-1	KOHLER	K-6426	KITCHEN SINK	WALL	1/2"	1/2"	2"	2"	UNDERMOUNT DOUBLE-BOWL LARGE/MEDIUM KITCHEN SINK	
WCO	ZURN	Z-1447	WALL CLEAN OUT	WALL	-	-	SEE REMARKS	-	MATCH CONNECTED PIPE SIZE UP TO 4". SECONDARY CLOSURE PLUG	
UB-1	SHARKBITE	SKU25099	WASHING MACHINE UTILITY BOX	-	1/2"	1/2"	2"	2"	UB BOX CAN BE USED ON COPPER,CPVC, PEX AND PE-RT. MAXI. WORKING PRESSURE: 200 PSI., MAXI. TEMPERATURE: 200°F, LEAD FREE DZR BRASS.	
M	SENSUS	WDS-10017-07	WATER METER	-	-	1 1/2"	-	-	NOMINAL DIAMETER (DN): 1 1/2", MAXIMUM OPERATING PRESSURE: 200 PSI, FLOW RATE: 0.5 - 200 GPM	
WCO-1	J.R. SMITH	4400 SERIES	WALL CLEAN OUT	WALL	-	-	SEE REMARKS	-	SPIGOT FERRULE WITH CAST BRONZE TAPER THREAD PLUG. CHROME PLATED BRONZE ROUND FRAME AND VANDAL PROOF SCREWS.	
FCO	J.R. SMITH	4020-U SERIES	FLOOR CLEAN OUT	FLOOR	-	-	SEE REMARKS	-	NO-HUB OUTLET, DUCO C.I. CLEANOUT WITH ROUND ADJUSTABLE ACORIATED NICKEL BRONZE ROUND TOP. VANDAL PROOF TOP.	
FD-1	J.R. SMITH	2005Y-A-B-.50	FLOOR DRAIN	FLOOR	-	-	2"	2"	DUCO C.I. BODY AND FLASHING COLLAR WITH SEDIMENT BUCKET, NICKEL BRONZE ADJUSTABLE STRAINER HEAD, ROUND GRATE, PROVIDE TRAP SEAL INSERTS OR 1/2" TRAP PRIMER CONNECTION.	
SH-1	PRAXIS	M 4836 SH	SHOWER HEAD	WALL	1/2"	1/2"	2"	2"	CONFIRM MODEL AND COLOR WITH ARCHITECTURE.	

NOTE:
1. REFER TO ARCHITECTURE, OWNERSHIP OR INTERIOR DESIGNER FOR ALL FIXTURE SELECTION AND ACCESSORIES. ALL ARE SUBJECT TO CHANGE.

EXPANSION TANK SCHEDULE

SYMBOL	QTY.	TANK VOLUME (GAL)	MAX. ACCEPTANCE FACTOR	TANK ACCEPT. VOLUME (GAL)	TYPE	BLADDER MATERIAL	SHELL MATERIAL	MAX. OPERATING TEMPERATURE (°F)	MAX. WORKING PRESSURE (PSIG)	SHIPPING WEIGHT (LBS)	DIMENSION (DIA. X HIEGHT)	MODEL	BASIS OF DESIGN	NOTES
ET-1	1	2.0	0.45	0.9	REPLACEBLE BLADDER	HEAVY DUCTY BUTYL	STEEL	200	150	5	8"X13"	ST-5	AMTROL	1,2,3

NOTES:
1. INSTALL PER MANUFACTURER INSTRUCTIONS.
2. PROVIDE WITH STANDARD SCHRADER TIRE VALVE CONNECTION.
3. ADJUST CHARGE PRESSURE IN FIELD.

GAS WATER HEATER SCHEDULE

SYMBOL	TYPE	QTY.	LOCATION	TANK CAPACITY (GAL)	RATED CAPACITY (GAL)	GAS INPUT THOUS. BTU/HR	TEMPERATURE RISE (°F)	RECOVERY IN G.P.H. 90°F RISE	WATER CONNECTION SIZE (INCH.)	SUPPLY GAS CONNECTION (INCH.)	VENT SIZE (INCH.)	ELECTRIC DATA		DIMENSION (INCH.)		SHIPPING WEIGHT (LBS)	UNIFORM ENERGY FACTOR (UEF)	MODEL	BASIS OF DESIGN
												FLA	V/PH/HZ	DIAMETER	HEIGHT				
GWH-1	PROPANE	1	GARAGE	40	38	36	90	36.3	3/4"	1/2"	4"Ø	5.0	120/1/60	19-1/2"	63-1/4"	135	0.66	XP40T06HE36U0	RHEEM

NOTES:
1. INSTALL PER MANUFACTURER INSTRUCTIONS.
2. USE ONLY VENT TERMINALS PROVIDED OR FACTORY AUTHORIZED TERMINALS FOR VENTING THIS WATER HEATER.
3. THIS WATER HEATER REQUIRES IT'S OWN SEPARATE VENTING SYSTEM. DO NOT CONNECT THE EXHAUST VENT INTO ANY EXITING VENT OR CHIMNEY.

RECIRCULATING PUMP SCHEDULE

SYMBOL	DESCRIPTION	LOCATION	QTY.	FLOW (GPM)	HEAD (FEET)	FLANGE SIZE NPT(IN)	ELECTRICAL					WEIGHT (LBS)	MODEL	BASIS OF DESIGN
							HP	VOLTAGE	PHASE	RPM	F.L.A (AMPS)			
RE-1	RECIRCULATING PUMP	GARAGE	1	5	8	3/4	1/12	120	1	1725	1.75	21	SERIES 100	BELL & GOSSETT

NOTES:
1. PROVIDE L-6007-A HONEYWELL, INC.OR APPROVED EQUAL, AQUAST INSTALLED IN HOT WATER CIRCULATING LINE ADJACENT TO PUMP TO STOP PUMP AT SET TEMPERATURE IN FIELD.

THERMOSTATIC MIXING VALVE

TAG NO.	DESCRIPTION	CONNECTION		MODEL	MANUFACTURER	COMMENTS
		COLD	HOT			
TMV-1	THERMOSTATIC MIXING VALVE (LOCAL)	3/8"	3/8"	LFUSG-B	WATTS	ASSE #1070
TMV-2	THERMOSTATIC MIXING VALVE (SHOWER)	1/2"	1/2"	BILTMORE SERIES 900	WATTS	ASSE #1016

UNIT : PLUMBING FIXTURE DEMAND TABULATION

FIXTURE	DESCRIPTION	OCCUPANCY	QTY.	DRAINAGE FIXTURE UNITS	SUB-TOTAL	LOAD VALUES IN WATER (EACH) SUPPLY FIXTURE UNITS (WSFU)			LOAD VALUES IN WATER (TOTAL) SUPPLY FIXTURE UNITS (WSFU)			REMARK
						COLD	HOT	TOTAL	COLD	HOT	TOTAL	
DW-1	DISHWASHING MACHINE (AUTOMATIC)	PRIVATE	1	2	2	-	1.5	1.5	-	1.5	1.5	
KS-1	KITCHEN SINK	PRIVATE	1	2	2	1.125	1.125	1.5	1.1	1.1	1.5	
SH-1	SHOWER HEAD	PRIVATE	1	2	2	1.5	1.5	2.0	1.5	1.5	2	
WM	WASHING MACHINE	PRIVATE	1	3	3	3.0	3.0	4.0	3.0	3.0	4	
L-1	LAVATORY	PRIVATE	6	1	6	0.8	0.8	1.0	4.5	4.5	6	
WC-1	WATER CLOSET, FLUSH TANK (1.6 GPF WATER CLOSET)	PRIVATE	4	3	12	2.5	-	2.5	10	-	10	
BT-1	BATHTUB	PRIVATE	2	2	4	3.0	3.0	4.0	6.0	6.0	8	
FD-1	FLOOR DRAIN	PRIVATE	1	2	2	0	0	0	0	0	0	
MB-1	MOP SINK	PUBLIC	1	3	3	1.125	1.125	1.5	1.125	1.125	1.5	
TOTALS					36	DFU			27.3	18.8	34.5	WSFU
DFU = DRAINAGE FIXTURE UNITS					EIGHTH	INCH SLOPE PER FOOT			17	14.3	22.2	GPM
WSFU = WATER SUPPLY FIXTURE UNITS					4"	DIAMETER OF PIPE (INCHES)			1 1/4"	1 1/4"	1 1/2"	INCHES REQ'D.

PIPING MATERIALS SCHEDULE

PLAN TAG	PLUMBING SYSTEM			SYSTEM MATERIAL AND FITTING SPECIFICATION
	DESCRIPTION	INSTALLATION	SIZES	
SW	SANITARY (SOIL) PIPING	SUSPENDED	2-1/2" AND SMALLER	SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
			3" AND LARGER	SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
		BURIED	3" AND LARGER	SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
V	VENT PIPING (ALL SYSTEMS)	SUSPENDED	2-1/2" AND SMALLER	SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
				SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
			3" AND LARGER	SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
				SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
		BURIED	2" AND LARGER	SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
				SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL
CW	DOMESTIC COLD WATER DISTRIBUTION	SUSPENDED	2" AND SMALLER	POLYETHYLENE CROSSLINK PIPE,PEX 5206 WITH FITTINGS.
			2 1/2" AND LARGER	COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22
		BURIED	2" AND SMALLER	POLYETHYLENE CROSSLINK PIPE,PEX 5206 WITH FITTINGS.
			2 1/2" AND LARGER	COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22
HW & HWR	DOMESTIC HOT WATER DISTRIBUTION	SUSPENDED	2" AND SMALLER	POLYETHYLENE CROSSLINK PIPE,PEX 5206 WITH FITTINGS.
			2 1/2" AND LARGER	COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22
		BURIED	2" AND SMALLER	POLYETHYLENE CROSSLINK PIPE,PEX 5206 WITH FITTINGS.
			2 1/2" AND LARGER	COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22

NOTE:
THE MATERIALS ARE SUBJECT TO CHANGE WITH APPROVAL OF ARCHITECTURE OR OWNER.

Project
RESIDENCE OF
JOE HUBBARD
3495
COUNTY ROAD P
GLENN COUNTY,CA

Project No.:
DATE:
DRAWN BY:
DESIGNED BY:

Sheet Title
PLUMBING SCHEDULES

☐ Preliminary Dwg.
☐ Bidding Doc.
☐ Construction Doc.

Sheet:
P5.0